

IPA ASSIGNMENT - 6

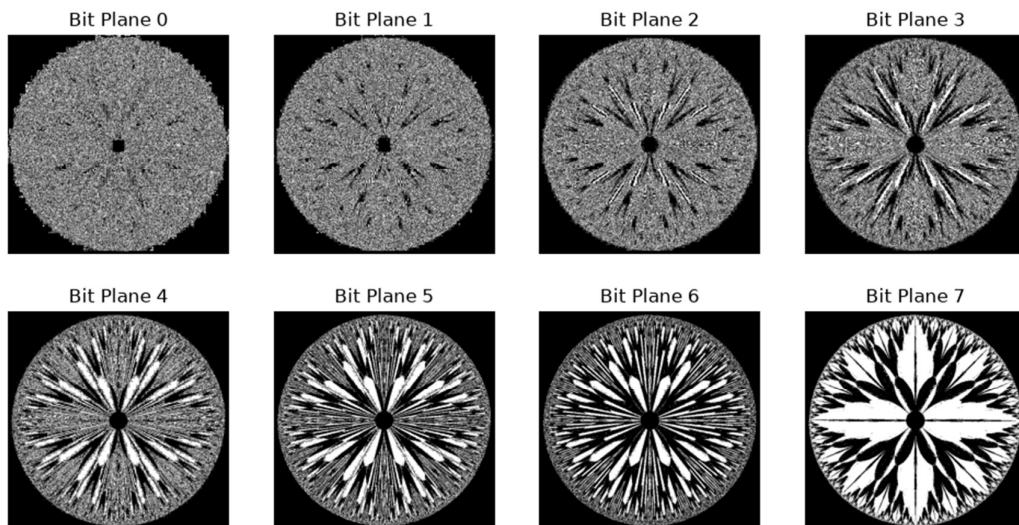
Name - Nishant Nahar

Roll No - 241551078

1. Bit Plane Slicing

```
import cv2
import matplotlib.pyplot as plt
img = cv2.imread("image.jpg", 0)
plt.figure(figsize=(12, 6))
for i in range(8):
    plane = (img >> i) & 1
    plt.subplot(2, 4, i + 1)
    plt.imshow(plane, cmap="gray")
    plt.title(f"Bit Plane {i}")
    plt.axis("off")
plt.show()
```

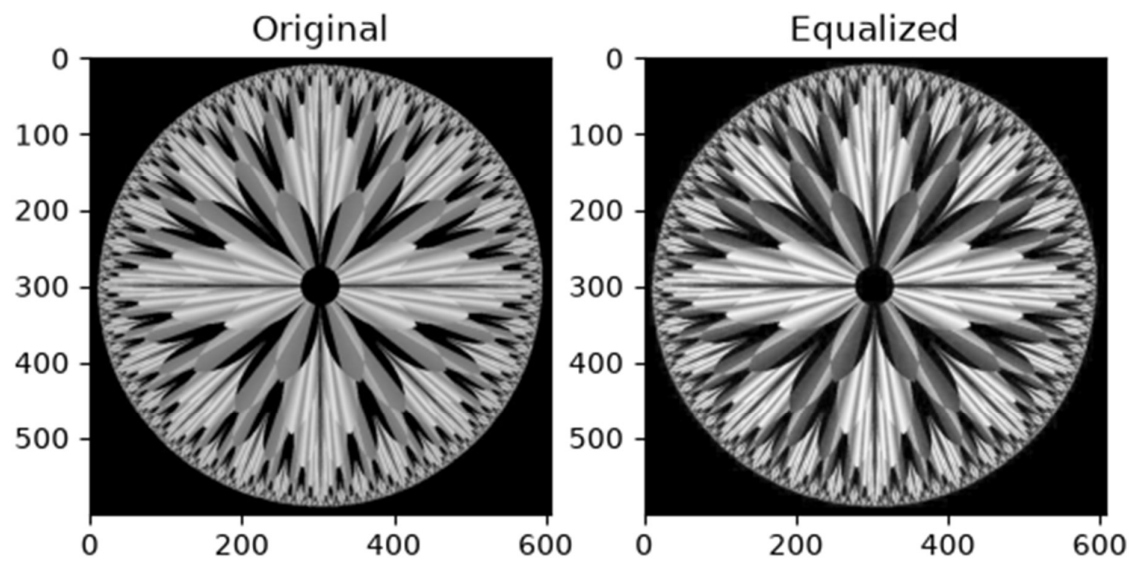
OUTPUT Image



2. Histogram Equalizations

```
import cv2
import matplotlib.pyplot as plt
img = cv2.imread("image.jpg", 0)
equalized = cv2.equalizeHist(img)
cv2.imwrite("equalized.jpg", equalized)
plt.subplot(121)
plt.imshow(img, cmap="gray")
plt.title("Original")
plt.subplot(122)
plt.imshow(equalized, cmap="gray")
plt.title("Equalized")
plt.show()
```

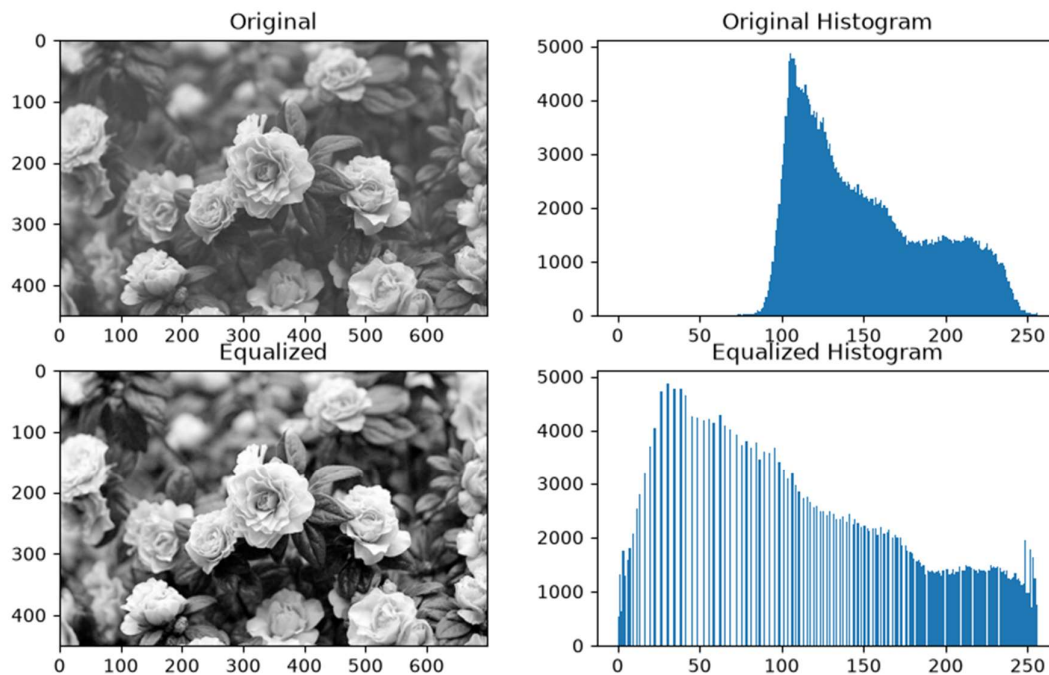
OUTPUT Image



3. Histogram of the Transformed and Original Image

```
import cv2
import matplotlib.pyplot as plt
img = cv2.imread("image_1.jpg", 0)
equalized = cv2.equalizeHist(img)
plt.figure(figsize=(10, 6))
plt.subplot(2, 2, 1)
plt.imshow(img, cmap="gray")
plt.title("Original")
plt.subplot(2, 2, 2)
plt.hist(img.ravel(), 256, [0, 256])
plt.title("Original Histogram")
plt.subplot(2, 2, 3)
plt.imshow(equalized, cmap="gray")
plt.title("Equalized")
plt.subplot(2, 2, 4)
plt.hist(equalized.ravel(), 256, [0, 256])
plt.title("Equalized Histogram")
plt.show()
```

OUTPUT Image



4. Blur Normal and Gaussian Blur

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
img = cv2.imread("image_2.jpg", 0)
average_kernel = np.ones((3, 3), np.float32) / 9
weighted_kernel = np.array([[1, 2, 1], [2, 4, 2], [1, 2, 1]], dtype=np.float32)/16
average = cv2.filter2D(img, -1, average_kernel)
weighted = cv2.filter2D(img, -1, weighted_kernel)
plt.figure(figsize=(12, 4))
plt.subplot(131)
plt.imshow(img, cmap="gray")
plt.title("Original")
plt.subplot(132)
plt.imshow(average, cmap="gray")
plt.title("Average Filter")
plt.subplot(133)
plt.imshow(weighted, cmap="gray")
plt.title("Weighted Filter")
plt.show()
```

OUTPUT Image

